Spatial–Spectral Weighted and Regularized Tensor Sparse Correlation Filter for Object Tracking in Hyperspectral Videos

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***Abstract*— Hyperspectral video camera captures spatial, spectral, and temporal information of moving objects. Traditional object tracking methods developed for color videos have been applied to hyperspectral videos after compressing hundreds of spectral bands into three, which does not fully use the wealth spectral information. To address this issue, we present a tensor sparse correlation filter (CF) with a spatial–spectral weighted regularizer for object tracking. First, tensor processing is used to reduce the spectral differences in homogeneous background, thereby producing robust spectral structure features. Second, a spatial–spectral weighted regularizer is designed in the CF framework to penalize filter template by suppressing the spectral features dissimilar to the center pixel in tracking. Third, a sparse constraint term and tracking context information are incorporated to suppress unexpected peaks in the response map. Finally, a reformulated stacked histogram of oriented gradient (HOG) feature extractor and a 2-D adaptive scale search strategy are developed to further improve the tracker's feature discrimination and scale adaptation capability. Experimental results demonstrate that the proposed method achieves superior tracking performance than the traditional CF-based trackers.**

***Index Terms*— Correlation filter (CF), hyperspectral video, object tracking, scale adaptive, tensor.**

I. Introduction

OBJECT tracking is a fundamental task of computer vision. It has been widely used in various applications such as video surveillance, robots, human–computer interaction, autonomous driving, and intelligent traffic control [1], [2], [3]. The traditional object tracking tasks rely on RGB or grayscale videos. Different from these common data modality, hyperspectral images (HSIs) provide fine spectral "diagnostic" information and store data in 3-D cubes with both the spatial and spectral dimensions. In HSIs, each material has unique spectral features, which offers potential for accurate object identification [4].

Recently, advances in electronics and sensor design have enabled the development of hyperspectral video cameras that can continuously capture HSIs at near video rate [5], [6], [7]. Hyperspectral video frames can simultaneously exploit the spatial, spectral, and temporal information of objects in the video. The spectral information provides details on the material constitution of target, which increases the interobject discrimination capability [8]. Therefore, hyperspectral videos have inherent advantages in real-time object detection and tracking tasks.

Hyperspectral object tracking aims to detect the size and locate the position of object in hyperspectral video frames. The object information is usually given in the first frame by a bounding box, and then the state of object in subsequent frames is predicted by the tracker [8], [9]. However, most tracking algorithms cannot be directly used for hyperspectral object tracking because they were developed for color videos. So far there are few publications on hyperspectral object tracking.

Banerjee et al. [10] first combined the traditional methods for HSI analysis and computer vision to achieve illumination-invariant motion detection and object tracking. Subsequently, considering radiation theory and particle filter, Van Nguyen et al. [11] proposed a framework to simultaneously track and identify an object based on its reflectance spectra. Using an adaptive hyperspectral sensor, Uzkent et al. [12] developed a hyperspectral likelihood map-aided tracking method (HLT) to build offline classifiers instead of learning a generative object model in an online manner for hyperspectral channels. Qian et al. [13] presented a convolutional network-based kernelized correlation filter (KCF) framework for the tracking objects, where a set of normalized 3-D cubes of object region was extracted as fixed convolution filters to contain spectral information surrounding an object.

In addition, considering the excellent performance of deep features in traditional object tracking, some deep learning methods were proposed for object tracking on hyperspectral video data. Uzkent et al. [14] proposed a deep hyperspectral KCF-based tracker (DeepHKCF) to efficiently track aerial vehicles, where KCFs and deep convolutional neural network features were applied. Liu et al. developed an anchor-free

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hyperspectral target tracking network (HA-Net), where the entire network is trained end-to-end, and high-level semantic features are exploited for target tracking [9]. Subsequently, a high spectral–spatial temporal resolution hyperspectral video object tracking framework (H^3Net) was proposed, where an end-to-end spectral classification branch was introduced to the anchor-free Siamese network, and the model was trained by the "cycle consistency" principle to increase the network's discriminative ability [15]. These deep-learning-based methods achieved superior tracking performance by linear or nonlinear transformation to obtain advanced features of the tracking objects. However, the performance of these trackers is limited due to small number of training samples.

Leveraging the advantage of correlation filter (CF)-based tracker that does not require training [16], [17] in object tracking, a CF-based fast spatial–spectral features (FSSFs) [18] method was proposed to implement robust, real-time object tracking in hyperspectral surveillance video, where a spatial–spectral convolution kernel was used to encode the discriminative information. To effectively extract the spatial–spectral feature directly from the mosaic spectral images, a feature descriptor using histogram of oriented mosaic gradient (HOMG) was proposed, where two forms of spectral filter arrays neighborhood (SFAN) were found to construct a feature extractor suitable for any spectral filter arrays structure [19]. To excavate material spatial–spectral information, a local spectral–spatial histogram of multidimensional gradients (SSHMG) and spatial distribution of materials method were proposed (abbreviated as MHT) [8] to describe the 3-D local spectral–spatial structure in hyperspectral video data. However, the MHT method has two major drawbacks. First, MHT takes background-aware CF (BACF) [20] tracker as the tracking benchmark. The BACF method was developed for color images, so the spectral information is ignored during BACF modeling. Although MHT used spectral information, since BACF was developed for color images, the spectral information is not sufficiently explored. Second, the abundance features from hyperspectral unmixing are used to describe detailed material distribution, where a priori spectrum library containing different materials is collected offline. When evaluating another dataset, a new spectral library needs to be retrained in advance. Although MHT automatically estimates the endmembers from the first frame of each video, the endmembers may change across frames, e.g., with reduced or added endmembers. Therefore, an effective without prior information method needs to be developed to explore the spectral properties of HSI, which helps accurately analyze objects in hyperspectral video tracking.

In this work, a tensor-based sparse CF with spatially and spectrally weighted and regularized (TSCFW) framework is proposed for hyperspectral object tracking task. To alleviate the spectral differences in objects in image frames, tensor processing is first used to obtain hyperspectral video datacube with fine spectral structure information. After that, with a predefined spatial–spectral weighted and regularized component, the CF model is reformulated to improve the discriminative ability of spectral information. Then a sparse response regularization term and context-aware information are added to the model to further enhance the discriminative ability of templates. Finally, a novel stacked histogram of oriented gradient (SHOG) feature extractor and a 2-D adaptive scale search strategy are used to enhance the robustness of features when dealing with the nonrigid deformation of the tracking object. The main contributions of this article are summarized as follows.

1) A novel hyperspectral object tracking framework is proposed, where tensor processing is used to reduce the spectral differences in homogeneous background, thereby producing robust spectral structure features. Simultaneously, sparse regularization term and context-aware information are used for discriminative CF to suppress unexpected peaks in the response map.
2) An effective spatial–spectral weighted regularizer is developed to suppress spectral features that are dissimilar to the center pixel of the tracking region. It improves the spectral discrimination ability of the template by measuring the spectral similarity and spatial distance between the center pixel and any neighborhood pixel.
3) A 2-D adaptive scale search strategy and a reformulated SHOG feature extractor are designed for the proposed model to improve the scale adaptive ability and the feature discriminative capability.

The remainder of this article is organized as follows. In Section II, a detailed description of the proposed framework is presented. In Section III, hyperspectral videos are used to verify the proposed method, and the experimental results and parameters are presented and discussed. Conclusion is drawn in Section IV.

II. Proposed Object Tracking Framework

Fig. 1 illustrates a flowchart of the proposed TSCFW framework, which consists of the following steps: 1) the tracking area and context-aware patches are determined according to the bounding box of the tracking object in the previous frame; 2) the tracking area with robust spectral structure characteristics is obtained by tensor processing; 3) the stacked SHOG features of tracking area and context-aware patches are designed to obtain robust representation features with spectral information; 4) these SHOG features are fed into the proposed TSCFW model to obtain a robust tracking template. Meanwhile, the adaptive scale search strategy is designed to obtain the optimal object scale; and 5) the position with the maximum response is used to locate the tracking object in the current frame.

A. Tracker Design

In the CF model, the desired filter template learns to regress the object region into a Gaussian distribution, which requires that the background is homogeneous. However, the appearance variance of the tracking object may produce many interference peaks, and its convolution with image is not conducive to highlight the center information and weaken the edge information [21]. By adding a sparse response constraint to the CF model, the undesired peak in the filter template is effectively suppressed, while the expected peak is highlighted, thereby

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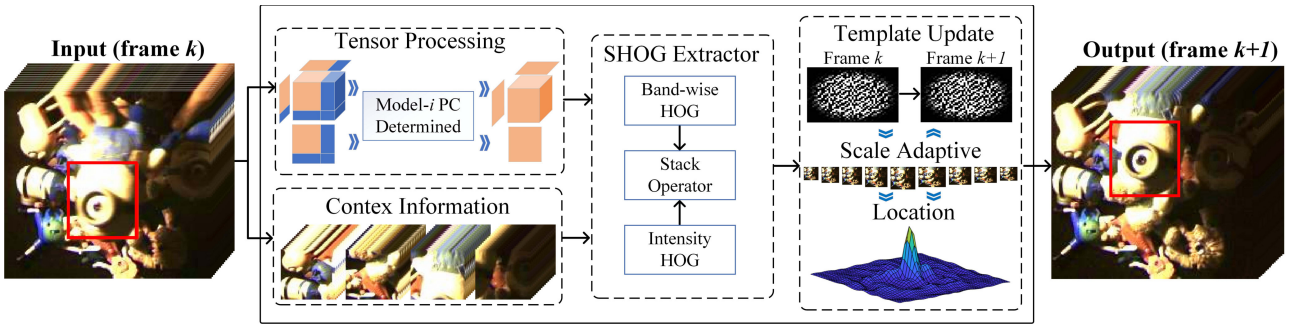


Fig. 1. Framework of the proposed TSCFW framework for hyperspectral object tracking.

improving the positioning accuracy. Furthermore, when the background is cluttered, the tracking performance is greatly affected by the surroundings of the tracked object. Therefore, adding contextual information to the filter during the learning stage can effectively improve the tracking performance.

Following the above idea, a sparse regularized CF with the spatial–spectral weighted and context-aware information model is proposed for hyperspectral object tracking. The new tracking framework is described as

$$\arg\min_{\boldsymbol{w}} f(\boldsymbol{w}, \tilde{\mathcal{X}}_{t_i}, \boldsymbol{y}) = \left\| \tilde{\mathcal{X}}_{t_i} \boldsymbol{w} - \boldsymbol{y} \right\|_2^2 + \lambda \|\boldsymbol{\Phi} \boldsymbol{w}\|_2^2 + \lambda_1 \|\tilde{\mathcal{X}}_{t_i} \boldsymbol{w}\|_2^2 + \lambda_2 \left\| \sum_{i=1}^{k} \boldsymbol{X}_i \boldsymbol{w} \right\|_2^2 \quad (1)$$

where $\tilde{\mathcal{X}}_{t_i} \in R^{I_1 \times I_2 \times I_3}$ is the newly reconstructed tensor datacube at time t_i, $\boldsymbol{y} \in R^{I_1 \times I_2}$ is the desired Gaussian-shaped response, and $\boldsymbol{w} \in R^{I_1 \times I_2}$ is the filter template. I_1, I_2, and I_3 represent the image rows, columns, and bands, which correspond to mode-1, mode-2, and mode-3 of the tensor, respectively. The first two terms form the CF function that is spatially and spectrally weighted, and the goal of its training is to find a solution that minimizes the squared error over sample $\tilde{\mathcal{X}}_{t_i}$ and its regression response $\boldsymbol{y}$. The third term is the sparse response term, which is the l_2-norm of the response used to suppress unexpected peaks in the response map. The fourth term is the context-aware constraint with k context patches outside of the object area. Four neighborhood context patches of the tracking object are used here for the training of the model, and these context patches $\boldsymbol{X_i} \in R^{I_1 \times I_2 \times I_3}$ are 3-D patches without tensor processing to speed up the operation. Note that although the object may not change much, its scale may change quickly with the movement, so the scale of these context patches changes across the moving objects. Hence, their size is set as the target size of the previous frame. λ, λ_1, and λ_2 are the regularization coefficients. $\boldsymbol{\Phi} \in R^{I_1 \times I_2}$ is the spatial–spectral weight matrix, which is specially designed for HSIs to explore the inherent spectral information. It is worth noting that in our proposed model, tensors cannot be considered as preprocessing, it is carried out together with the whole model, and the reconstructed patch image is related to the region location of the tracking object. The spatial–spectral weight matrix $\boldsymbol{\Phi}$ is with Gaussian normal hypothesis, which is expressed by

$$\Phi(\boldsymbol{\xi}, \boldsymbol{x}) = \exp\left(-\frac{\mathrm{d}_{\mathrm{spat}}(\boldsymbol{\xi}, \boldsymbol{x})}{2\sigma_d}\right) \cdot \exp\left(-\frac{\mathrm{d}_{\mathrm{spec}}(\boldsymbol{\xi}, \boldsymbol{x})}{2\sigma_e}\right) \quad (2)$$

where the first term is the spatial weight, the second term is the spectral weight, and $\boldsymbol{x}$ and $\boldsymbol{\xi}$ represent the center pixel vector (i, j) and any neighborhood pixel vector (k, l) in the predicted patch image, respectively. σ_d and σ_e are the constants used to control the outputting sensitivity. The spatial distance term $d_{\mathrm{spat}}(\boldsymbol{\xi}, \boldsymbol{x})$ and spectral distance term $d_{\mathrm{spec}}(\boldsymbol{\xi}, \boldsymbol{x})$ are calculated as

$$\mathrm{d}_{\mathrm{spat}}(\boldsymbol{\xi}, \boldsymbol{x}) = \sqrt{(i-k)^2 + (j-l)^2} \quad (3)$$

$$\mathrm{d}_{\mathrm{spec}}(\boldsymbol{\xi}, \boldsymbol{x}) = \|\boldsymbol{\xi} - \boldsymbol{x}\|^2 \quad (4)$$

where (i, j) and (k, l) are the spatial coordinates of $\boldsymbol{x}$ and $\boldsymbol{\xi}$, respectively. For further simplification, constants σ_d and σ_e in (2) are set to 1. After deduction, the spatial–spectral weight is expressed as

$$\phi = \exp\left(-\frac{1}{2}\left(\sqrt{(i-k)^2 + (j-l)^2} + \|\boldsymbol{\xi} - \boldsymbol{x}\|^2\right)\right) \quad (5)$$

where ϕ is a single element of matrix $\boldsymbol{\Phi}$ at position (k, l).

In (1), to modify the spatial location of the correct peak response and further enhance the spectral discrimination ability of the model, a spatial–spectral weight matrix is used to penalize the filter coefficients. In the response, the sparse correlation term is forced to learn a sparse response to decrease the influence of unexpected corruptions, because the filter template is expected to be Gaussian-shaped to easily locate the tracking center. In addition, the local context is also taken into account in the learned filter to effectively limit drift and the boundary effects. With the new objective function, the solution is derived as follows.

1) Solution in the Primal Domain: In the primal domain, objective function (1) is difficult to solve directly by derivation, since the filters in the first, third, and last terms of the tracking objective function aim to achieve different responses. The optimization problem can be reformulated as

$$f(\boldsymbol{w}, \mathcal{B}, \overline{\boldsymbol{y}}) = \|\mathcal{B}\boldsymbol{w} - \overline{\boldsymbol{y}}\|_2^2 + \lambda \|\boldsymbol{\Phi}\boldsymbol{w}\|_2^2 \quad (6)$$

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where the new tensor $\mathcal{B}$ matrix is represented as

$$\mathcal{B} = \begin{bmatrix} \tilde{\mathcal{X}}_{t_i} \\ \sqrt{\lambda_1}\tilde{\mathcal{X}}_{t_i} \\ \sqrt{\lambda_2}\boldsymbol{X}_1 \\ \vdots \\ \sqrt{\lambda_2}\boldsymbol{X}_k \end{bmatrix}, \quad \overline{\boldsymbol{y}} = \begin{bmatrix} \boldsymbol{y} \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}. \tag{7}$$

By deriving $\boldsymbol{w}$ in (6), and forcing it to be zero, the closed-form solution is obtained as

$$\boldsymbol{w} = (\mathcal{B}^T\mathcal{B} + \lambda\boldsymbol{\Phi}^T\boldsymbol{\Phi})^{-1}\mathcal{B}^T\overline{\boldsymbol{y}}. \tag{8}$$

Given (7), the closed-form solution of $\boldsymbol{w}$ is reformulated as

$$\boldsymbol{w} = \frac{\tilde{\mathcal{X}}_{t_i}^T\boldsymbol{y}}{(1+\lambda_1)\tilde{\mathcal{X}}_{t_i}^T\tilde{\mathcal{X}}_{t_i} + \lambda_2\sum_{i=1}^{k}\boldsymbol{X}_i^T\boldsymbol{X}_i + \lambda\boldsymbol{\Phi}^T\boldsymbol{\Phi}}. \tag{9}$$

Inspired by previous works [17], [21], (9) is usually learned in the Fourier domain for efficient training and testing. Thus, the final closed-form solution of $\boldsymbol{w}$ is converted into the Fourier domain

$$\hat{\boldsymbol{w}} = \frac{\hat{\boldsymbol{x}}^* \odot \hat{\boldsymbol{y}}}{(1+\lambda_1)\odot\hat{\tilde{\boldsymbol{x}}}_{t_i}^* \odot \hat{\tilde{\boldsymbol{x}}}_{t_i} + \lambda_2\sum_{i=1}^{k}\hat{\boldsymbol{x}}_i^* \odot \hat{\boldsymbol{x}}_i + \lambda\hat{\boldsymbol{\Phi}}^* \odot \hat{\boldsymbol{\Phi}}} \tag{10}$$

where $\odot$ denotes the elementwise product, $*$ is complex conjugate, and $\hat{}$ denotes the discrete Fourier transform operation $\mathrm{fft}(\cdot)$ [i.e., $\hat{\boldsymbol{x}} = \mathrm{fft}(\boldsymbol{x})$]. For a testing patch image $\hat{\boldsymbol{z}}$, the final response is

$$\hat{\boldsymbol{R}}_p = \hat{\boldsymbol{z}} \odot \hat{\boldsymbol{w}}. \tag{11}$$

2) Solution in the Dual Domain: Regardless of the convexity of the optimization function in the primal domain, its dual function is always a convex optimization problem. Therefore, when there is no optimal solution in the primal domain, the dual-domain solution is replaced as the optimal solution.

Equation (6) is solved in the dual domain. For a given dual variable $\boldsymbol{\alpha}$, the filter template $\boldsymbol{w}$ is represented by $\boldsymbol{w} = \mathcal{B}^T\boldsymbol{\alpha}$. Given (8), the closed-form solution of $\boldsymbol{\alpha}$ is calculated by

$$\boldsymbol{\alpha} = (\mathcal{B}\mathcal{B}^T + \lambda\boldsymbol{\Phi}\boldsymbol{\Phi}^T)^{-1}\overline{\boldsymbol{y}}. \tag{12}$$

Considering (7), in the Fourier domain, the solution of $\boldsymbol{\alpha}$ is denoted by

$$\hat{\boldsymbol{\alpha}} = \left(\begin{bmatrix} \boldsymbol{d}_{0,0} & \boldsymbol{d}_{0,1} & \boldsymbol{d}_{0,2} & \cdots & \boldsymbol{d}_{0,k+2} \\ \boldsymbol{d}_{1,0} & \boldsymbol{d}_{1,1} & \boldsymbol{d}_{1,2} & \cdots & \boldsymbol{d}_{1,k+2} \\ \boldsymbol{d}_{2,0} & \boldsymbol{d}_{2,1} & \boldsymbol{d}_{2,2} & \cdots & \boldsymbol{d}_{2,k+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{d}_{k+2,0} & \boldsymbol{d}_{k+2,1} & \boldsymbol{d}_{k+2,2} & \cdots & \boldsymbol{d}_{k+2,k+2} \end{bmatrix}\right)^{-1}\begin{bmatrix} \hat{\boldsymbol{y}} \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \tag{13}$$

where vectors $\boldsymbol{d}_{j,l}$ with $j, l \in \{0, 1, \ldots, k+2\}$ are given by

$$\begin{cases} \boldsymbol{d}_{0,0} = \mathrm{diag}\left(\hat{\tilde{\boldsymbol{x}}} \odot \hat{\tilde{\boldsymbol{x}}}^* + \lambda\boldsymbol{\phi} \odot \boldsymbol{\phi}^T\right) \\ \boldsymbol{d}_{1,1} = \mathrm{diag}\left(\lambda_1\hat{\tilde{\boldsymbol{x}}} \odot \hat{\tilde{\boldsymbol{x}}}^* + \lambda\boldsymbol{\phi} \odot \boldsymbol{\phi}^T\right) \\ \boldsymbol{d}_{0,1} = \boldsymbol{d}_{1,0} = \mathrm{diag}\left(\sqrt{\lambda_1}\hat{\tilde{\boldsymbol{x}}} \odot \hat{\tilde{\boldsymbol{x}}}^* + \lambda\boldsymbol{\phi} \odot \boldsymbol{\phi}^T\right) \\ \boldsymbol{d}_{j,l} = \mathrm{diag}\left(\lambda_2\left(\hat{\boldsymbol{x}}_j \odot \hat{\boldsymbol{x}}_l^*\right) + \lambda\boldsymbol{\phi} \odot \boldsymbol{\phi}^T\right), \\ \qquad\qquad j, \quad l \geq 2 \\ \boldsymbol{d}_{j,l} = \mathrm{diag}\left(\sqrt{\lambda_2}\left(\hat{\tilde{\boldsymbol{x}}}_j \odot \hat{\boldsymbol{x}}_l^*\right) + \lambda\boldsymbol{\phi} \odot \boldsymbol{\phi}^T\right), \\ \qquad\qquad j = 0, \quad l \geq 2 \text{ or } j \geq 2, \ l = 0 \\ \boldsymbol{d}_{j,l} = \mathrm{diag}\left(\sqrt{\lambda_1\lambda_2}\left(\hat{\tilde{\boldsymbol{x}}}_j \odot \hat{\boldsymbol{x}}_l^*\right) + \lambda\boldsymbol{\phi} \odot \boldsymbol{\phi}^T\right), \\ \qquad\qquad j = 1, \quad l \geq 2 \text{ or } j \geq 2, \ l = 1. \end{cases} \tag{14}$$

Referring to [22], in the dual domain, the detection response of a testing patch image $\boldsymbol{z}$ is reformulated as

$$\hat{r}_p = \boldsymbol{z}\mathcal{B}^T\boldsymbol{\alpha} = \hat{\boldsymbol{z}} \odot \hat{\tilde{\boldsymbol{x}}}^* \odot \hat{\boldsymbol{\alpha}}_0 + \sqrt{\lambda_1}\hat{\boldsymbol{z}} \odot \hat{\tilde{\boldsymbol{x}}}^* \odot \hat{\boldsymbol{\alpha}}_1 + \sqrt{\lambda_2}\sum_{i=1}^{k}\hat{\boldsymbol{z}} \odot \hat{\boldsymbol{x}}^* \odot \hat{\boldsymbol{\alpha}}_{i+2}. \tag{15}$$

3) Online Update: During tracking, as the appearance changes, the difference between the initial object and the current object becomes greater, resulting in an inferior performance. By online update strategy, a reasonable filter template is found to improve the performance of the tracker. Therefore, to account for object changes (e.g., scale and pose changes during tracking), the filter template $\boldsymbol{w}$ is updated by an autoregressive method with a learning rate ζ

$$\boldsymbol{w}_t = (1-\zeta)\boldsymbol{w}_{t-1} + \zeta\boldsymbol{w}_t \tag{16}$$

where the filter template $\boldsymbol{w}$ is computed by (10).

B. Tensor Processing

The background variation in the tracking object caused by various factors is one of the main challenges of the hyperspectral object tracking task. Solely using spatial texture information of object cannot effectively identify the tracking object in subsequent frames. Hence, it is crucial to adopt a more effective method to cater for the structural characteristics of hyperspectral data. Recently, some tensor-based work [23], [24], [25], [26] has shown potential to effectively process the 3-D hyperspectral data. Considering the spectral similarity of tracking target in different hyperspectral frames, tensor representation is adopted to incorporate 3-D spatial structure information of spectra. Meanwhile, to obtain robust features, the Tucker decomposition [27], [28], [29], which can well exploit the 3-D spectral and spatial information of hyperspectral data, is used to extract high-order principal component (PC) information from multitemporal sequence images.

The third-order tensor better approximates the real hyperspectral data structure. Therefore, each frame of hyperspectral video is denoted as a third-order tensor, which is formulated as $\boldsymbol{\mathcal{X}}_{t_i} \in R^{I_1\times I_2\times I_3}$, where $\boldsymbol{\mathcal{X}}_{t_i}$ is an HSI at time t_i. Due to the influence of various factors, including scale variation

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(SV), fast motion, occlusion, etc., the spectral curves of a tracking object may change in multitemporal sequences. To obtain more robust features, the Tucker decomposition is used to reconstruct these sequence images. The tensor $\boldsymbol{\mathcal{X}}_{t_i}$ is represented as

$$\boldsymbol{\mathcal{X}}_{t_i} \approx \mathcal{G} \times_1 \boldsymbol{U} \times_2 \boldsymbol{V} \times_3 \boldsymbol{W} \tag{17}$$

where $\mathcal{G} \in R^{I_1 \times I_2 \times I_3}$ is the core tensor and its elements represent the level of interaction between distinct components. $\boldsymbol{U} \in R^{I_1 \times I_1}$, $\boldsymbol{V} \in R^{I_2 \times I_2}$, and $\boldsymbol{W} \in R^{I_3 \times I_3}$ are three mutually orthogonal factor matrices. The optimization problem is stated as

$$\begin{aligned} &\arg\max_{\boldsymbol{U},\boldsymbol{V},\boldsymbol{W}} \|\boldsymbol{\mathcal{X}}_{t_i} \times_1 \boldsymbol{U} \times_2 \boldsymbol{V} \times_3 \boldsymbol{W}\|_2^F \\ &\text{s.t.} \begin{cases} \mathcal{G} \in R^{I_1 \times I_2 \times I_3} \\ \boldsymbol{U} \in R^{I_1 \times I_1}, \quad \boldsymbol{V} \in R^{I_2 \times I_2}, \ \boldsymbol{W} \in R^{I_3 \times I_3} \\ \boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{I}_1, \quad \boldsymbol{V}^T\boldsymbol{V} = \boldsymbol{I}_2, \ \boldsymbol{W}^T\boldsymbol{W} = \boldsymbol{I}_3. \end{cases} \end{aligned} \tag{18}$$

In (18), the alternating least squares (ALSs) algorithm is used to obtain the core tensor and factor matrices [**?**], [27].

By the Tucker decomposition, a third-order tensor matrix is decomposed into three factor matrices ($\boldsymbol{U}, \boldsymbol{V}, \boldsymbol{W}$) and a core tensor $\mathcal{G}$, where significant information is obtained by truncating suitable r PCs. Inspired by [28], an energy-cumulative method is used to determine the numbers of PCs in three factor matrices, which is

$$\eta = \frac{\sum_{i=1}^{r} a_i}{\sum_{i=1}^{m} a_i} \tag{19}$$

where $a_i = [a_1, a_2, \ldots, a_m]$ is the singular value of factor matrix, m is the number of all the nonzero singular values, and r is the selected PC numbers. By setting the value of η, the corresponding r in different factor matrices is automatically determined, where η is set empirically. Generally, the more complex the background, the smaller the rate η. In our method, the parameter η is fixed at 0.95 to facilitate subsequent processing tasks.

The reconstructed $\tilde{\mathcal{X}}_{t_i}$ has the same spectral dimension as the $\mathcal{X}_{t_i}$, which is obtained by

$$\tilde{\mathcal{X}}_{t_i} \approx \mathcal{G}_r \times_1 \boldsymbol{U}_{r_1} \times_2 \boldsymbol{V}_{r_2} \times_3 \boldsymbol{W}_{r_3} \tag{20}$$

where $\mathcal{G}_r = \mathcal{G}(1:r_1, 1:r_2, 1:r_3)$, $\boldsymbol{U}_r = \boldsymbol{U}_r(:, 1:r_1)$, $\boldsymbol{V}_r = \boldsymbol{V}_r(:, 1:r_2)$, and $\boldsymbol{W}_r = \boldsymbol{W}_r(:, 1:r_3)$. r_i is the PC number of different factor matrixes.

Completing the reconstruction operation, a reconstructed tensor $\tilde{\mathcal{X}}_{t_i} \in R^{I_1 \times I_2 \times I_3}$ is obtained. After the same processing of hyperspectral frames is completed, the newly reconstructed video frames $\tilde{\mathcal{X}}_{t_1}, \tilde{\mathcal{X}}_{t_2}, \ldots, \tilde{\mathcal{X}}_{t_n}$ corresponding to the scene at time $t_1, t_2, \ldots, t_n$, respectively, are used for object tracking.

Note that in our proposed TSCFW model, the tensor processing cannot be simply regarded as preprocessing, because it is constructed together with the model. The position and size of tensor processing in each frame are determined by the previous frame, rather than performing tracking task after tensor processing. Using this operation, the spectral structural information of the tracking object is effectively preserved, which makes the tracking spectral signal consistent in different frames.

C. Feature Extraction

Feature extraction and object discrimination are two main tasks in object tracking. In the traditional CF-based methods, grayscale or hand-crafted features of background patches are used to learn filter. As a 3-D cube, HSI comprises two spatial dimensions and one spectral dimension. However, traditional grayscale or histograms of oriented gradients (HOGs) [30] features are only for 2-D images, which is not suitable for multiband HSI. Previous works show that SSHMG features are effective for hyperspectral object tracking [8]. Moreover, bandwise HOG (BHOG) [14] features, which are constructed by concatenating the HOG feature across all the bands, can also significantly improve the tracking performance in hyperspectral videos.

The choices of feature extractors are important for the proposed TSCFW tracker. The appropriate feature extractor can significantly improve the tracking performance. Different from the SSHMG features, considering the orthogonal relationship between the spectral dimension and the spatial dimension, the stacked HOG (SHOG) features are redefined to extract discriminative features, which stack the HOG features of all the bands to fully use multibands of HSI. Since the spectral response range of object is acquired in different bands, the characteristics of each band are unique. The SHOG features can more effectively maintain the independence of features between different bands. In addition, the intensity of HSI represents the spectral reflection ability of various objects. Therefore, the intensity HOG features are also extracted and stacked together to form the SHOG features. Note that SHOG is different from BHOG, which constructed by concatenating the BHOG features and intensity HOG features. When extracting the HOG features of each band, it was found that the HOG features of the last dimension are all zero values, so the truncated HOG features are used for stacking operation in SHOG. The reformulated SHOG extractor is constructed as follows:

$$\begin{cases} F(X)_{\text{shog}} = \text{stack}\left(\sum_{i=1}^{I_3} \Theta\big(\text{hog}\big(\tilde{\mathcal{X}}_i\big)\big), \Theta\big(\text{hog}\big(I\big(\tilde{\mathcal{X}}\big)\big)\big)\right) \\ I\big(\tilde{\mathcal{X}}\big) = \sum_{i=1}^{I_3} \big(\tilde{\mathcal{X}}_i\big) \end{cases} \tag{21}$$

where $\text{hog}(\cdot)$ represents the HOG feature extraction operations. $\Theta(\cdot)$ is the truncation operation, which means to remove the last dimension. $\text{stack}(\cdot)$ is the stack operation, which stacks different features together. I_3 is the number of bands. Besides, the intensity of HSI is also important. Therefore, in the SHOG feature, the intensity HOG is also extracted and stacked thereafter. $I(\tilde{\mathcal{X}})$ represents the intensity image, which is calculated by superposing radiance values band by band.

D. Adaptive 2-D Scale Space Search

In object tracking task, when dealing with the nonrigid deformation of object, trackers which use the 1-D scale estimation method often lose the tracking object [31], [32].

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Fig. 2. Examples of changes in terms of the aspect ratio. Frames are selected from the *book* scene in the hyperspectral video dataset. (a) Frame #0001:1.208. (b) Frame #0289:1.688. (c) Frame #0595:1.231.

To effectively deal with the SVs in video sequences, an adaptive 2-D scale space search method is used [32].

Suppose that the original object's size is $P \times R$. The 1-D scale space search method [33], [34] is denoted as $a^n P \times a^n R$, where a denotes the scale factor, and $n \in \{\lfloor -((S-1)/2) \rfloor, \ldots, \lfloor ((S-1)/2) \rfloor\}$ is the scale parameters. S represents the number of scales. In this scale estimation method, the aspect ratio is fixed. In fact, the variance of width and height is not the same as each other, when facing severe nonrigid deformation, and the appearance variance of the tracking object cannot be covered.

As shown in Fig. 2, in the *book* scene, the ground-truth bounding boxes of the tracking object in raw frames are given, where the shape change in the book is nonproportional in different frames. In the #0001, #0289, and #0595 frames, their width and height are $W_1 = 24$, $H_1 = 29$, $W_2 = 16$, $H_2 = 27$, and $W_3 = 26$, $H_3 = 32$, respectively. Then, their aspect ratios are: $a_1 = H_1/W_1 = 1.208$, $a_2 = H_2/W_2 = 1.688$, and $a_3 = H_3/W_3 = 1.231$, respectively. Comparing the aspect ratio, it is observed that the appearance variance of the tracking object is not proportional. Hence, an identical scale factor cannot cater to the actual object tracking.

To effectively deal with the appearance variance, a 2-D adaptive scale space search method is used, denoted as $a^m P \times a^n R$. The 2-D scale parameters $m, n \in \{\lfloor -((S-1)/2) \rfloor, \ldots, \lfloor ((S-1)/2) \rfloor\}$ allow the scale factors to change along two axis directions, aiming to capture severe nonrigid deformation of the tracking object. By combining different scales along two axis directions, $S \times S$ image patches with different aspect ratios are obtained. Therefore, for a testing patch image $\boldsymbol{z}$, its scale set is obtained as

$$\boldsymbol{Z} = \{\boldsymbol{z}_1, \ldots, \boldsymbol{z}_i, \ldots, \boldsymbol{z}_{S\times S}\}, \quad i \in \{1, 2, \ldots, S \times S\}. \tag{22}$$

In the scale space, the response score of different sizes is calculated using (11). Note that the size of $\boldsymbol{z}_i$ is different, which needs to be resized to the fixed object size $P \times R$ before subsequent CF is applied. Then, the maximum response is used to find its corresponding scale s_{opt}. s_{opt} is the most suitable object scale, which is calculated by

$$s_{\text{opt}} = \arg\max_i \ \text{ifft}(\hat{\boldsymbol{z}}_i \odot \hat{\boldsymbol{w}}) \tag{23}$$

where ifft$(\cdot)$ is the inverse Fourier transform operation. Note that the purpose of (23) is to obtain an appropriate object scale without positioning and updating. Since tensor processing is time-consuming, the original datacube without tensor processing is used to obtain the appropriate object scale to reduce the computational cost. After obtaining s_{opt}, the final response R_p is calculated by

$$R_p = \text{ifft}(\hat{\boldsymbol{z}}_{s_{\text{opt}}} \odot \hat{\boldsymbol{w}}) \tag{24}$$

where R_p is used to locate the position of tracking object in the current frame.

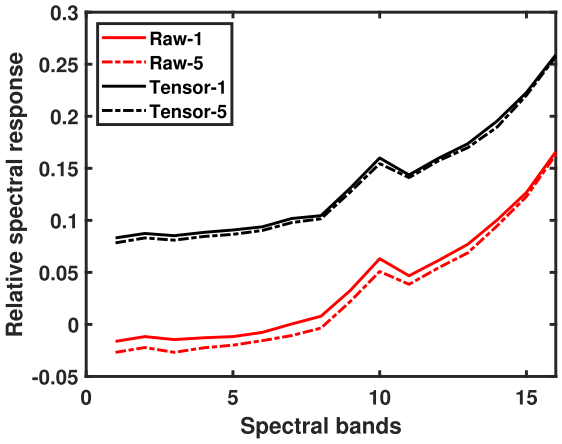
E. Spectral Analysis After Tensor Processing

Tensor processing can effectively retain the spectral information. Simultaneously, other impure signal is eliminated, which is essential for sequential hyperspectral frames. Using this operation, the raw hyperspectral frame is compressed to preserve r_i PCs for different factor matrices, which makes the tracking of spectral signals to have less fluctuation in different frames. To verify the performance of tensor processing, spectral angle is used as a metric to analyze these differences after tensor processing of hyperspectral video dataset.

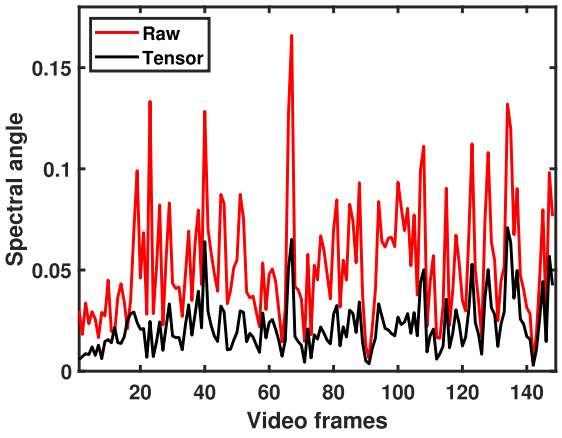
In Fig. 3, we take *coin* scene as an example to illustrate the difference between the raw spectrum and the new spectrum after tensor processing. The raw spectral pair is selected from the center region of the tracking object in the first and fifth frames, respectively. Here, the average of pixels in a 3×3 window of the tracking center is used to replace a single pixel for comparison. After tensor processing, the raw pixel pair and the reconstructed pixel pair are plotted in Fig. 3(a), respectively. It is observed that the trend of the red and black solid lines or the red and black dotted lines remains consistent, which indicates that tensor processing can preserve the spectral information. However, the spectrum of the tracking object in continuous frames is slightly different. The raw spectral difference is slightly larger than the difference after tensor processing. In Table I, the spectral angles of two pairs of pixels are listed. The spectral angle becomes smaller when the raw pixels' pair is projected to the tensor domain, which further verifies that tensor processing can significantly reduce the spectral fluctuation of the tracking object.

As shown in Fig. 3(b), the spectral angles change in all the consecutive frames. Similarly, the spectral pairs are also selected from the 3×3 window surrounding the center pixel of the tracking object in successive frames, and the average of

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(a)



(b)

Fig. 3. Illustration of spectral change in *coin* scene. (a) Average spectrum change in the first and fifth frames. (b) Average spectral angles' change in all the successive frames.

TABLE I
AVERAGE SPECTRAL ANGLE VALUES UNDER DIFFERENT DOMAINS (NORMALIZED)

Scenes	Raw images	Tensor reconstructed images
coin	0.079	0.014

pixels is used for comparison. From the visualization curves, the fluctuation of the black curve is obviously smaller than that of the red curve. It indicates the spectral differences in consecutive frames are significantly reduced after tensor processing, which is more beneficial to object tracking in subsequent hyperspectral video frames.

III. EXPERIMENTAL RESULTS AND DISCUSSION

To demonstrate the superiority of the proposed TSCFW method, several typical CF trackers are taken as the baseline methods. These include circulant structure of tracking-by-detection with kernels (CSK) [35], KCF [17], scale adaptive with multiple features tracker (SAMF) [33], fast discriminative scale space tracker (fDSST) [34], and BACF [20]. Extensive experiments are conducted on a benchmark hyperspectral/false-color video datasets [8]. In our experiments, the model's parameters were set as $\eta = 0.95$, $\lambda = 10$, $\lambda_1 = 1$, and $\lambda_2 = 0.1$. The scale parameters were set as $S = 3$ and $a = 1.05$. The learning rate was set as $\zeta = 0.0019$. The parameters of SHOG feature followed the suggestion in MHT [8], where the cell size and orientations were set to 4×4 and 9 respectively. All the other parameters were set as the same as those in BACF [20].

All the experiments were performed in a desktop computer. The machine has Windows 10 with a 64-bit operating system, an Intel Core i7-8700H CPU with 3.20 GHz and 28 GB of main memory, and an NVIDIA GeForce GTX 1660Ti GPU. The code was implemented in MATLAB R2018a.

A. Datasets' Description

To validate the performance of the proposed method, a baseline hyperspectral video dataset was used to perform hyperspectral object tracking. This dataset was provided by Xiong et al. [8] in hyperspectral object tracking challenge competition, which was acquired using an XIMEA snapshot camera. These videos were captured at 25 frames/s. Each frame was originally captured in 2-D with 16 bands arranged in a mosaic mode. The spectral range is from 470 to 620 nm, and each band image consists of 512×256 pixels. These shared videos have been preprocessed, including spectral calibration and image registration. In our experiments, all 35 test video scenes were used to verify the proposed method.

In these video scenes, all the videos were also labeled with associated challenging factors, including in-plane rotation (IPR), out-of-plane rotation (OPR), background clutters (BCs), occlusion (OCC), SV, deformation (DEF), motion blur (MB), low resolution (LR), illumination variation (IV), fast motion (FM), and out-of-view (OV). In Table II, these basic attributes are listed. More detailed description can be found in [8] and [15].

B. Evaluation Metrics

To evaluate the performance of various trackers, three metrics including precision plot, success plot, and area under curve (AUC) were used to make evaluation [8], [18]. All the results were recorded with one-pass evaluation (OPE).

In the precision plot, the precision rate represents the percentage of frames where the center location error (CLE) is less than the given threshold. The CLE denotes the center difference between the predicted bounding box by the tracker and the ground truth, which is defined as

$$\mathrm{CLE} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \tag{25}$$

where (x_1, y_1) and (x_2, y_2) represent the coordinates of the central location corresponding to the predicted bounding box and the actual ground truth, respectively. The precision rate is defined as

$$\mathrm{PR}(\zeta) = \frac{N_{\mathrm{CLE}<\zeta}}{N_{\mathrm{total}}} \tag{26}$$

where ζ is a given threshold and its value is an integer from 0 to 50 pixels. When CLE $< \zeta$, a frame is considered as

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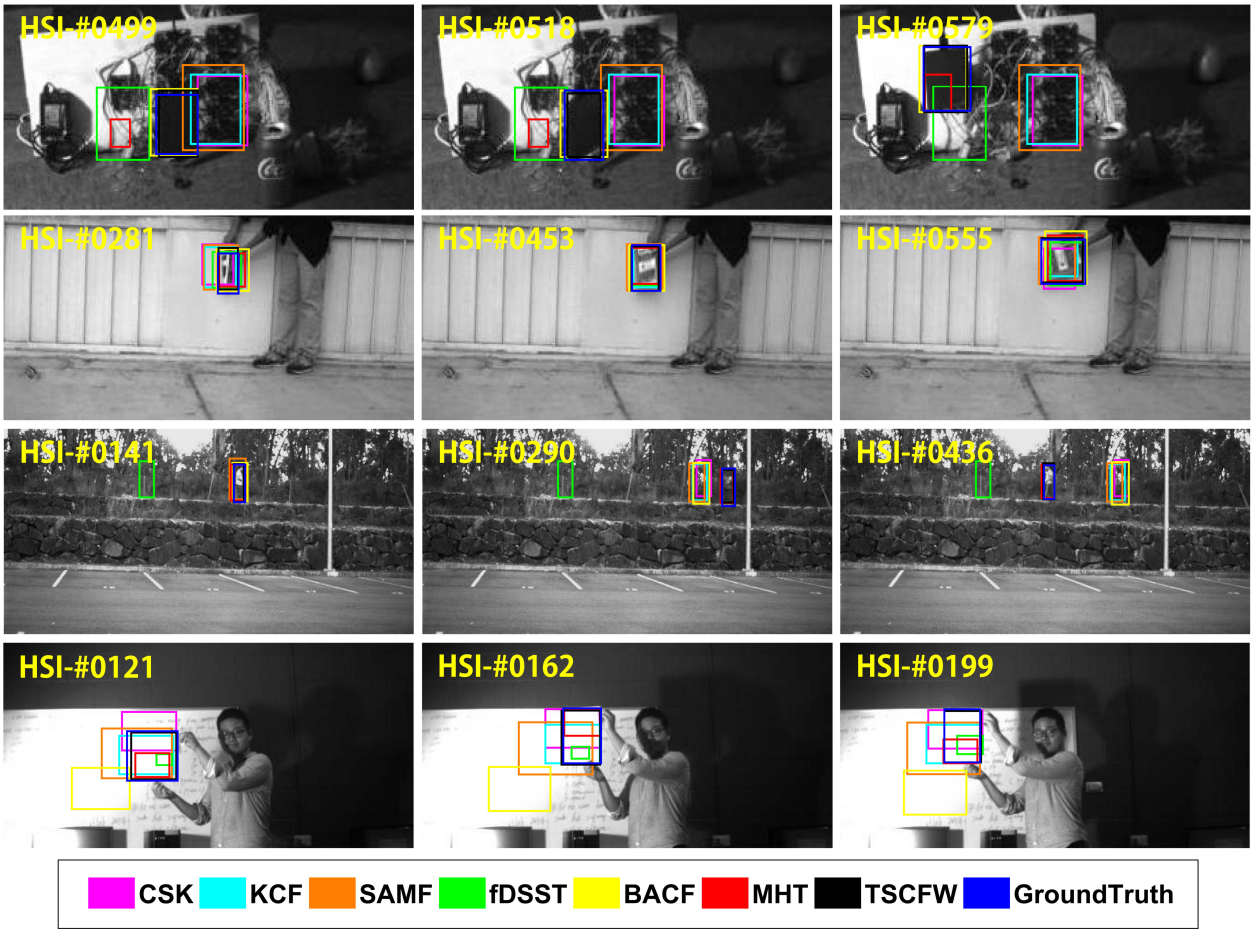


Fig. 4. Visualization of results on four video sequences (i.e., *drive, book, forest, and paper*). Bounding boxes in different colors are the results from different trackers.

a successfully tracked frame. $N_{\mathrm{CLE}<\zeta}$ represents the number of frames with CLE less than ζ, and N_{total} is the number of total frames.

In success plot, the success rate is used to measure the intersection over union (IOU) between the predicted bounding box and the actual ground truth. The IOU is defined as

$$\mathrm{IOU} = \frac{B_t \cap B_p}{B_t \cup B_p} \tag{27}$$

where B_t and B_p represent the area of the ground truth and the predicted bounding box, respectively. $\cap$ and $\cup$ represent the intersection and the union operations of two regions, respectively. The success rate is calculated as

$$\mathrm{SR}(\tau) = \frac{N_{\mathrm{IOU}>\tau}}{N_{\mathrm{total}}} \tag{28}$$

where τ is a threshold denoting different overlaps between the predicted bounding box and the ground truth, with a value from 0 to 1. When the IOU is larger than τ, the frame is considered as a successful frame.

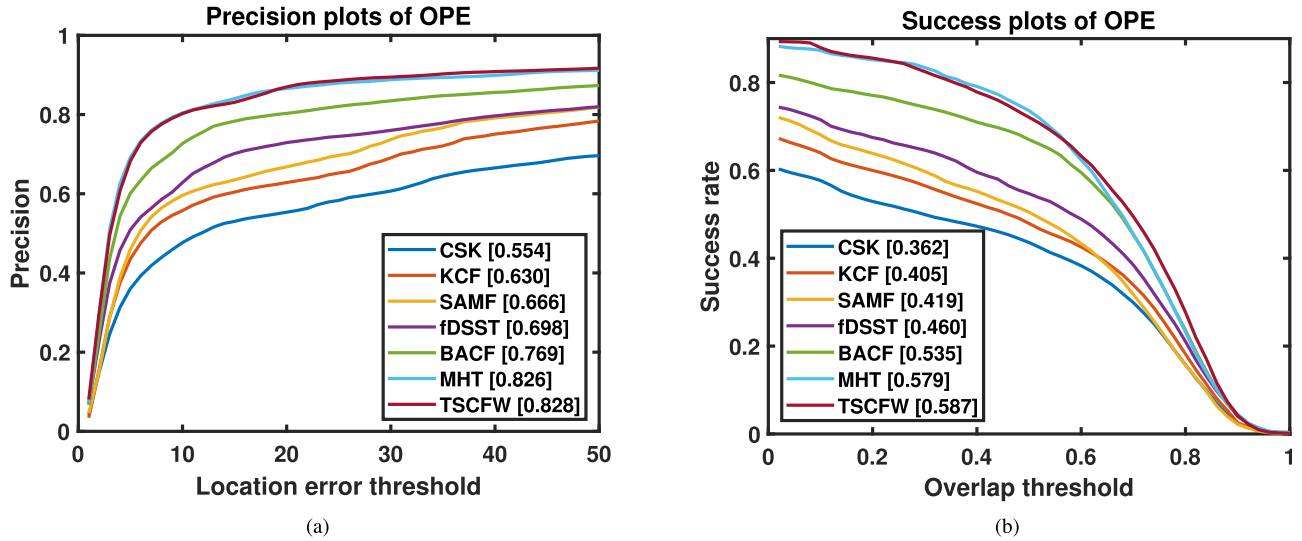
Taking CLE and PR(ζ) as the x-axis and y-axis, respectively, the precision plot is obtained, where the threshold of CLE varies from 1 to 50. Similarly, taking overlap threshold τ and $SR(\tau)$ as the x-axis and y-axis, respectively, the success plot is obtained. The AUC of each success plot is also selected as an evaluation criterion to rank all the trackers. Intuitively, the trackers are evaluated by the AUC values with a larger value denoting a better one.

C. Results' Comparison

To evaluate the performance of the proposed tracker, some CF object tracking methods are used for competitive comparison. Notably, the proposed TSCFW and MHT were tested on hyperspectral videos, while CSK, KCF, SAMF, fDSST, and BACF were developed for color images and were run on the corresponding false-color videos. Fig. 4 shows the visualization results of all the trackers on *drive*, *book*, *forest*, and *paper* scenes. Fig. 5 presents the tracking results of all the competing trackers on all the video scenes, where the precision plots, success plots, and the corresponding AUC values are used to qualitatively and quantitatively evaluate the tracking performance of different trackers.

The tracking results of various methods are displayed in Fig. 4. In the visualization results, TSCFW outperforms all the others in the tracking performance, especially when it encounters severe nonrigid deformation, such as in *drive*, *book*, and *paper* video scenes. In Fig. 5, the performance of various trackers in all the scenes is also evaluated through precision plots and success plots shown in Fig. 5(a) and (b), respectively.

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(a) (b)

Fig. 5. Comparison of various trackers. AUCs of these trackers are specified in the legends. (a) Precision plots of OPE. (b) Success plots of OPE.

TABLE II
BASIC PROPERTIES OF HYPERSPECTRAL VIDEO SCENES

Scenes	Frames	Size	Attributes
ball	625	471×207×16	SV,MB,OCC
basketball	186	463×256×16	FM,MB,OCC,LR
board	471	277×140×16	IPR,OPR,BC,OCC,SV
book	601	317×148×16	IPR,OPR,DEF
bus	131	261×148×16	LR,BC,FM
bus2	326	361×167×16	IV,SV,OCC,FM
campus	976	384×157×16	IV,SV,OCC
car	101	261×148×16	SV,OCC,IPR,OPR
car2	131	351×167×16	SV,IPR,OPR
car3	331	512×256×16	SV,LR,OCC,IV
card	930	363×200×16	IPR,BC,OCC
coin	149	291×120×16	BC
coke	731	493×207×16	BC,IPR,OPR,FM,SV
drive	725	297×142×16	BC,IPR,OPR,SV
excavator	501	500×240×16	IPR,OPR,SV,OCC,DEF
face	279	446×224×16	IPR,OPR,SV,MB
face2	1111	446×224×16	IPR,OPR,SV,OCC
forest	530	512×256×16	BC,OCC
forest2	363	512×256×16	BC,OCC
fruit	552	493×232×16	BC,OCC
hand	184	314×186×16	BC,SV,DEF,OPR
kangaroo	117	385×206×16	BC,SV,DEF,OPR,MB
paper	278	446×224×16	IPR,BC
pedestrain	306	351×167×16	IV,SV
pedestrain2	363	512×256×16	OCC,LR,DEF,IV
player	901	463×256×16	IPR,OPR,DEF,SV
playground	800	463×256×16	SV,OCC
rider1	336	512×256×16	LR,OCC,IV,SV
rider2	210	512×256×16	LR,OCC,IV,SV
rubik	526	493×207×16	DEF,IPR,OPR
student	396	438×256×16	IV,SV
toy1	376	271×135×16	BC,OCC
toy2	601	371×171×16	BC,OCC,SV,IV,OPR
truck	221	512×256×16	OCC,IV,SV,OV
worker	1209	228×121×16	SV,LR,BC

In the precision plots, the closer the curve is to the upper left corner, the better the tracking performance. On the contrary, in the success plots, the closer the curve is to the upper right corner, the better the tracking performance. The results show that CSK gives inferior precision due to the fact that it only learns a template by dense sampling. When the background is similar to the tracking object, the feature of the tracking object is disturbed, resulting in tracking failure. In contrast, other CF-based trackers, such as KCF, SAMF, fDSST, and BACF, obtain more negative samples and robust object features for template training. Therefore, better tracking performance is obtained.

Taking BACF as the tracking benchmark, MHT's performance is significantly better than BACF. The reason is that MHT uses the SSHMG feature extractor to obtain object's spatial–spectral features. In addition, the abundance information of material is also taken into account in the model. Therefore, when facing with slight occlusion, IV, etc., the detection performance of MHT is effectively stimulated, resulting in more stable tracking effects. In Fig. 5, the proposed TSCFW shows much better tracking performance than other trackers. In the TSCFW model, the context-aware information and spatial–spectral features of tracking object are considered. Meanwhile, tensor representation is also used to obtain more robust spectral structure information, which is more helpful to obtain discriminative features to learn a robust template, thereby significantly improving the prediction of object location.

Table III lists the AUC values of success plots, distance precision when CLE is less than 20 pixels (DP@20), and frames/s. These quantitative results also confirm that the proposed TSCFW achieves the most appealing performance. Compared with other trackers, the execution cost of TSCFW and MHT is still high, especially for MHT. The reason is due to that the feature extraction of hyperspectral videos with more channels is time-consuming. In addition, tensor processing is also computationally expensive in the proposed TSCFW model.

Furthermore, the tracking performance under different hyperspectral video attributes is analyzed to better highlight the advantages of the proposed TSCFW tracker. Note that in our experiment, all the hyperspectral videos containing the same challenging factor are used for attribute evaluation.

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TABLE III

OVERALL PERFORMANCE COMPARISON OF VARIOUS TRACKERS IN HYPERSPECTRAL/FALSE-COLOR VIDEOS. THE BEST AND SECOND BEST VALUES ARE HIGHLIGHTED BY RED AND BLUE FONTS, RESPECTIVELY

Methods	CSK [35]*	KCF [17]*	SAMF [33]*	fDSST [34]*	BACF [20]*	MHT [8]	TSCFW
AUC	0.362	0.405	0.419	0.460	0.535	**0.579**	**0.587**
DP@20	0.553	0.628	0.668	0.729	0.803	**0.866**	**0.870**
FPS	**662.749**	**291.496**	53.175	138.277	43.442	2.244	3.412

* denotes the corresponding false-color videos were used for evaluation.

TABLE IV

ATTRIBUTE-BASED COMPARISON ON HYPERSPECTRAL/FALSE-COLOR VIDEOS. THE BEST AND SECOND BEST VALUES ARE HIGHLIGHTED BY RED AND BLUE FONTS, RESPECTIVELY. OUR TRACKER RANKS THE FIRST ON 5 OUT OF 11 ATTRIBUTES, i.e., SV, LR, IPR, BC, AND OV, AND THE SECOND BEST IN OTHER ATTRIBUTES

Attributes	CSK [35]*	KCF [17]*	SAMF [33]*	fDSST [34]*	BACF [20]*	MHT [8]	TSCFW
SV	0.352	0.358	0.376	0.463	**0.551**	0.550	**0.587**
MB	0.303	0.456	0.529	0.574	**0.668**	0.605	**0.613**
OCC	0.329	0.388	0.411	0.453	0.526	**0.567**	**0.546**
FM	0.310	0.364	0.453	**0.611**	0.551	0.525	**0.583**
LR	0.043	0.071	0.143	0.337	0.340	**0.493**	**0.533**
IPR	0.554	0.655	0.629	0.619	0.674	**0.680**	**0.728**
OPR	0.553	0.581	0.581	0.578	**0.714**	0.633	**0.701**
DEF	0.566	0.557	0.577	0.602	0.671	**0.682**	**0.672**
BC	0.144	0.360	0.701	**0.705**	0.651	0.691	**0.775**
IV	0.213	0.135	0.162	0.325	**0.488**	0.384	**0.404**
OV	0.051	0.321	0.301	0.470	0.543	**0.626**	**0.654**

* denotes the corresponding false-color videos is used for evaluation.

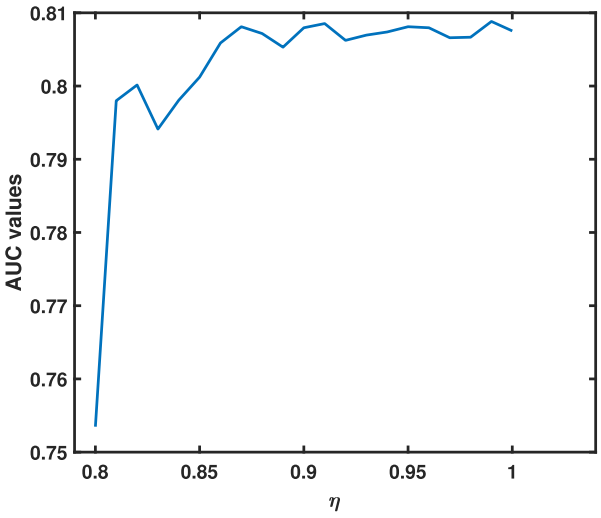


Fig. 6. AUC values of success plot under different tensor rates η on *face* scene.

TABLE V

ABLATION STUDY ON WEIGHT MATRIX AND TENSOR PROCESSING (AUC%)

Method	AUC	ΔAUC	DP@20	ΔDP@20	FPS
both	58.7%	-	87.0%	-	3.412
with weight matrix	56.9%	-1.8%	83.8%	-3.2%	4.289
with tensor	51.8%	-6.9%	77.5%	-9.5%	3.438
neither	51.7%	-7.0%	76.7%	-10.3%	4.649

TABLE VI

ABLATION STUDY ON CONTEXT-AWARE INFORMATION (AUC%)

Method		AUC	DP@20	FPS
with context	without tensor	58.7%	87.0%	3.412
	with tensor	56.1%	84.7%	1.759
without context		54.4%	81.8%	4.665

Table IV lists the AUC values for these trackers in different challenges. It is observed that in SV, LR, IPR, BC, and OV attributes, TSCFW achieves the best tracking performance. When encountering other attributes, the second best tracking results are obtained. The results further verify the superior performance of the proposed TSCFW in hyperspectral video object tracking.

D. Ablation Analysis

As shown in Fig. 6, the impact of tensor processing on the tracker is verified. Taking *face* scene as example, by changing the value of η from 0.8 to 1 with an interval of 0.01, the tracking performance with tensor processing is collected. It is observed that the tracking performance is different when η takes various values. When η reaches 0.87, the tracking performance tends to be stable, and when $\eta = 1$, that is, when tensor processing is not implemented, the tracking precision does not reach the peak, which further indicates that selecting the appropriate η for tensor processing can effectively improve the tracking performance of the TSCFW tracker. In practice, to reduce parameters as much as possible, η is set to 0.95 as an empirical value.

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The ablation study results on weight matrix and tensor processing are listed in Table V. The results show that the tracker has the most inferior performance when the model does not carry out tensor processing and spatial–spectral weighted operations. When adding one of the operations, the tracking ability of the model is improved by a certain extent. Obviously, when both the operations are implemented, the proposed TSCFW produces the optimal tracking performance. The proposed TSCFW has a 7% improvement in the AUC of success plot and 10.3% improvement in DP@20. This is because after adding spatial–spectral weighted matrix, the model has better ability to distinguish spectral information. After that, the tracking ability of the model is significantly improved by adding tensor processing, which further verifies that tensor processing and spatial–spectral weight matrix significantly improve the tracking performance in hyperspectral videos.

In terms of execution time, when tensor processing is not implemented, the AUC values of the proposed TSCFW decrease. However, its efficiency is improved, which meets the needs of practical application. Since SHOG is a 3-D extension of the HOG features, the processing is more time-consuming, which is the main reason for reduced tracking efficiency.

Furthermore, the influence of context-awareness on tracking performance is also discussed. In Table VI, the results with and without context-awareness are listed. It is observed that the tracker with context information outperforms the tracker without information. These results verify that adding context information to the model can significantly improve the robustness of the filter template. After the context patches are processed by the tensor, the tracking performance and tracking speed of TSCFW are reduced, because these context patches are used as negative samples to train the template. After the tensor operation is performed, the spectral difference between negative samples and tracking object is reduced, resulting in the decrease in tracking performance. Therefore, during model building, these 3-D context patches are not processed by tensor to speed up operation, which caters to rapid processing in application.

IV. Conclusion

In this article, a novel tensor-based sparse CF with spatial–spectral weight matrix and context-aware frame has been introduced for hyperspectral object tracking. The tensor processing can reconstruct robust video datacubes by truncating noises. A novel spatial–spectral weighted matrix is added to the filter template to increase the spectral discrimination capability. Meanwhile, spares regularization term and context-aware information are also taken into account in the model to restrain unexpected peaks in the response map. Finally, SHOG features and 2-D adaptive scale search strategy are used to further enhance the feature discrimination and scale adaptation ability. The experimental results have confirmed that the proposed method outperforms several tracking methods based on CFs. In the future, we will further improve the computational cost of the proposed method for the requirement of real applications.

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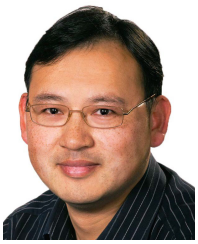
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